



Investigation report

Haul truck and light vehicle collision, Bindarra Gold Mine - NTSB-style probable cause report

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INVESTIGATION WORKFLOW SUITE

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Executive summary

What happened

At 02:47:12 on 12/08/2026, an empty 220 t class rigid haul truck was descending the main ramp of the Stage 4 open pit at Bindarra Gold Mine. A light vehicle came out of the RL 312 dewatering spur and turned right across the descending lane. The truck braked from 37 km/h and struck the light vehicle's right side at 24 km/h. It pushed the vehicle 18.4 m down the ramp and rolled it onto its left side against the inner windrow.

The sole occupant was a grade control technician employed by a survey subcontractor, on his third night shift at the mine. He sustained a fractured left clavicle, three fractured left ribs and a small pneumothorax, and was conscious throughout. His seatbelt was fastened and loaded. The roll-over protection structure deformed without being breached. He was extricated through the rear of the vehicle 27 minutes after the collision and flown to the metropolitan trauma centre at 07:40. The light vehicle was destroyed. The truck was damaged at the front left and returned to service on 14/08/2026. The Stage 4 pit stood down for 9 hours 15 minutes, and the main ramp below RL 340 was closed for 22 hours.

What the investigation found

The spur was opened under a works order that excluded it from management of change. Nothing was therefore referred to the traffic management plan's owner, the survey department or the contractor's technical services group. The spur opened with 41 m of sight distance against a 96 m stopping distance at the posted speed. It was not on the positive-call list, which covered nine intersections and could change only at the annual review. The light vehicle driver slowed and looked before the truck was in view, then drove out without stopping. Neither driver knew of the other until six seconds before impact. These factors very likely contributed.

Other factors likely contributed. The intersection had no stop control, warning sign or delineation. The driver's access card was issued without a check for the pit driving assessment. Neither driver made a radio call, which the plan did not require at an unlisted intersection. A collision alert suppression geofence covered the spur without operators' knowledge. It contributed through the operator's expectation that a quiet system meant an empty ramp, not through the alert itself. The state guideline set no review trigger and no basis for the positive-call list. That finding is held at the standard of proof and no higher. It concerns the width of the guideline's discretion, not any failure by the regulator, and the mine was not in breach.

The truck operator braked, applied the retarder and steered away from the pit edge within 1.4 seconds of the beacon coming into view. This reduced the impact speed and kept the light vehicle against the inner windrow.

Two safety issues did not contribute. These were a haul roster of fourteen consecutive 12 hour 30 minute night shifts, and a night shift extrication that waited about 9 minutes for a second responder called from the village. It could not be determined whether airborne dust delayed the operator's sighting of the beacon, or whether the operator was fatigued.

What can be done

A new road into an operating pit changes the traffic network, whatever document creates it. Controls that depend on a list, a drawing or a geofence boundary need a route by which that change reaches them between reviews.

At 05:10 that morning, the Mining Manager directed a positive call before entering any Stage 4 intersection, and the RL 312 spur was closed. Survey measured the other ten unlisted roads cut since the last review. Two below the plan's sight distance rule were closed pending a stop control. The contractor's technical services group added suppressed proximity events to its monthly collision awareness report.

The investigation recommended that BIN-TMP-004 provide a route onto the positive-call list between annual reviews and require new roads to be referred to the plan's owner. It recommended that ORR-PRO-022 stop excluding earthworks that create a haul road intersection, or carry the management of change referrals on the works order. It also recommended that a geofence boundary change be treated as a change to a documented control, with operators told in writing where alerts are suppressed. No action had been recorded against the guideline, the access card check, the roster or the night shift extrication arrangement.

Factual information

History of the occurrence

The RL 312 dewatering spur is a 240 m access road to dewatering bore DW-14 in the Stage 4 pit at Bindarra Gold Mine. It was cut under works order WO-26-4471, raised by the Mining Engineer on 28/07/2026, approved by the Mining Superintendent on 29/07/2026 and completed on 31/07/2026.

On 02/08/2026 the bore was energised and light vehicles first used the spur. The "ROAD CLOSED - CONSTRUCTION" sign at the spur mouth was removed that day and not replaced.

At about 22:10 on 04/08/2026, a dewatering technician drove out of the spur in front of a descending haul truck, then reversed and let it pass. At 23:48 he submitted hazard report HAZ-2026-0917, which read in part: "you cannot see up the ramp from that road. There is a rise in the ramp and a windrow along the side and you are out on the road before you can see anything." The Dewatering Superintendent closed it on 06/08/2026, citing a reminder at the 05/08 pre-start to take care and give way to haul trucks, and reduced the risk rating from Medium to Low. The report was not referred to the mining department, the contractor, the survey department or the traffic management plan owner.

On 09/08/2026 the light vehicle driver, a grade control technician employed by Hallorn Survey & Geotechnical, was issued a site access card at the Bindarra gatehouse. He began his first night shift that evening.

At 02:22:41 on 12/08/2026, dispatch returned the Stage 4 crews from crib. HT-407 completed one cycle. At 02:46:21 its operator called the three-sixty crossover, empty and descending.

The light vehicle driver finished a set of levels at the DW-14 collar and drove LV-118 out along the spur. His departure is estimated at about 02:45, from the spur length and his position at the collision. His own recollection, about ten to three, is later than the recorded collision time. In an interview that had not been finalised, he said he slowed at the mouth, looked right up the ramp, saw nothing and turned right to go up. He said he was travelling at about 25 km/h and did not believe he came to a complete stop. The turn took LV-118 across the descending lane, right side towards a descending truck.

At 02:47:06.0 HT-407 was travelling at 37 km/h at RL 317, 53 m above the spur mouth. The survey places this as the first point at which the light vehicle's beacon was visible over the crest. At 02:47:07.4, 38.5 m above the mouth, the service brakes and retarder were applied and the truck was steered 14 degrees right. At 02:47:12.0 the truck's front left struck the right side of LV-118 at 24 km/h, 6.2 m into the ramp from the spur mouth. LV-118 was pushed 18.4 m down the ramp, went onto its left side 11.2 m from the impact and came to rest against the inner windrow. HT-407 stopped at 02:47:19, 19 m below the intersection.

Neither driver made a positive radio call before entering the intersection.

The operator reported the collision on the pit channel at 02:47:41. At 02:48:01 dispatch stopped all Stage 4 units. The operator was held in his cab until 02:48:44, then went to the light vehicle driver. The emergency response team was paged at 02:48. The first response vehicle entered the pit at 02:51 and reached the scene at 02:54. The paramedic completed the primary survey and extrication plan at 02:58. The second responder arrived from the accommodation village at 03:07 with the hydraulic rescue tool. The driver was extricated through the rear of the vehicle at 03:14, 27 minutes after the collision. He reached the surface medical facility at 03:41 and was flown to the metropolitan trauma centre at 07:40. The occurrence was notified at 03:26 under ORR-OCC-2026-0332.

Injuries and damage

The light vehicle driver sustained a fractured left clavicle, three fractured left ribs and a small pneumothorax. He was conscious throughout.

LV-118 was destroyed. HT-407 was damaged at the front left bumper, left mudguard and lower two access ladder steps, and returned to service on 14/08/2026. The Stage 4 pit stood down for 9 hours 15 minutes. The main ramp below RL 340 was closed for 22 hours.

Personnel information

The haul truck operator had 11 years in open pit mining, 7 of them on 220 t class rigid haul trucks. He had worked 14 months at Bindarra and 4 years 2 months with Tarrowen Mining Services. His verification of competency on type was issued in 06/2025 and his annual refresher completed in 02/2026. He had received no traffic management refresher since his site general induction in 06/2025. He was on night 10 of a 14 night swing, 8 hours 47 minutes into a 12 hour 30 minute shift, and had slept 6 hours 30 minutes in the preceding day. He described himself as "Tired, the way you are on night ten. Not falling asleep tired." His drug and alcohol test was negative.

The light vehicle driver had 11 years in survey, 6 in mining grade control, and 4 years with Hallorn. He completed Hallorn's two hour online general mining induction on 24/07/2026. His Bindarra card followed a 20 minute gatehouse briefing on muster points, speed limits and personal protective equipment. No document showed completion of the Bindarra site-specific supplement or the light vehicle pit driving assessment. He was on the third night of a 9 on 5 off swing, working 18:00 to 06:00. No duty or rest concerns were identified.

The night shift supervisor had 19 years in open pit mining and 2 years 6 months in the role. He was also on night 10 of 14 and was at the ROM pad at the time. His drug and alcohol test was negative.

Equipment information

HT-407 is a 220 t class rigid haul truck. Its monthly brake test on 03/08/2026 recorded 0.80 m/s² net deceleration. Its retarder and service brakes performed as specified during the occurrence.

Its collision awareness system operated at Level 8 (operator advisory). Level 9 (automated intervention) is available on the fleet but not enabled at Bindarra. An alert is raised when a tag enters a 60 m ring at a closing rate above 10 km/h, but not at any range inside a suppression geofence.

The in-cab FMS map for Stage 4, last published on 24/07/2026, did not show the RL 312 spur.

LV-118 carried a passive CAS tag and no FMS. Its 4.5 m amber beacon whip was sheared at the base, and the lamp worked when tested after recovery. The roll-over protection structure was deformed inward at the right rear mounting but not breached. The driver's seatbelt was fastened, and the webbing showed loading marks.

Meteorological information

The ramp surface was dry. No measurement of airborne dust was available. The haul truck operator said: "There was some hanging around, yeah. I couldn't tell you if it made a difference." The light vehicle driver, in his unfinalised interview, said the air was clear and there was no dust in the spur.

Recorded data

The FMS download (TRW-TS-2026-114) recorded 1.4 seconds between the last steady data point and the first brake pressure rise. Mean deceleration from brake application to collision was 0.78 m/s² net of the grade, over 4.6 seconds and 39 m.

The CAS event log for HT-407 recorded:

- "18:47:09.200 PROXIMITY HT-407 tag LV-118 range 21.4 m closing 32 km/h zone S4-LOAD alert SUPPRESSED (geofence)" [CAS event log, HT-407, UTC]

No alert was presented in the cab. Outside a geofence, an alert would have been raised at about 02:47:06.5.

The pit channel was recorded continuously throughout.

Site information

At RL 312 the main ramp runs approximately north-west on a 9.1 per cent descending grade, 24 m wide between windrows. Traffic keeps left. The spur joins on a descending truck's left, on the inside of a left-hand bend, at an included angle of 71 degrees, 34 m below a convex grade break that crews call the crest.

From a seated eye height of 5.80 m, a descending operator first sees the spur mouth at 41 m. The top of a 4.5 m beacon whip on a vehicle in the spur is visible from 53 m, but the vehicle body is not. From an eye height of 1.45 m at the spur mouth, a driver first sees a descending truck at 38 m. A 1.6 m inner windrow obscures the truck's lower body over the last 25 m of the spur approach.

The stopping distance of an empty 220 t class truck at the posted 40 km/h on a 9 per cent descent is 96 m.

Organisational and management information

The Bindarra traffic management plan (BIN-TMP-004) requires:

- sight distance at every intersection of not less than the stopping distance of the largest vehicle, and never less than 120 m (clause 6.2)
- a warning sign 50 m ahead on each approach, a give-way control and delineation at intersections shown on the plan's road layout drawing (clause 6.4).

The RL 312 spur was not on that drawing, nor among the nine intersections listed for positive radio calls. The operator, the supervisor and the light vehicle driver each described calls being made only at those nine.

The plan was reviewed on 14/11/2023, 10/11/2024 and 12/11/2025. No out-of-cycle review under clause 9.1 has occurred in its eleven year history. Of 14 roads cut in Stage 4 since the last review, 11 were cut under works orders and 3 under management of change records. Only the 3 were referred to the plan owner.

WO-26-4471 classified the works as routine and recorded management of change as not required under ORR-PRO-022 clause 5.3. It contained no signage, survey, plan referral, FMS or geofence scope. Its handover field read "N/A".

The S4-LOAD suppression geofence was drawn on 14/03/2025 to stop nuisance alerts at the loading face. The contractor's technical services group extended it northward on 09/06/2026, bringing the boundary to 180 m up-ramp of the spur. The size of the extension is [NOT ESTABLISHED: previous boundary of the S4-LOAD geofence]. The extension was not reviewed against the traffic management plan. No site training covered suppression geofences, and the standing alert statistics report excludes suppressions.

The haul truck operator said he had never had an alert in the pit in 14 months and did not know of the suppression. The supervisor had known of the suppression at the dig since mid 2025 but not of the June 2026 extension.

Contractor Management Standard ORR-STD-009 clause 7.4 states that the site supplement and pit driving assessment "shall not be substituted". Clause 7.5 makes the engaging department responsible for verifying them before a card is issued. The gatehouse issues cards against the recognition agreement alone. No permit, call-in, board or pit entry check identified a light vehicle entering an operating pit.

The night shift is rostered with one paramedic. The second responder required before any extrication is called from the village, 11 km from the gatehouse, in the vehicle carrying the hydraulic rescue tool. The post-incident review found an approximately 9 minute wait for the second responder and recorded that it did not affect the clinical outcome.

Additional information

A search of Bindarra hazard reports for the 24 months to 11/08/2026 returned three besides HAZ-2026-0917, from 03/2025, 09/2025 and 02/2026. Each described a light vehicle and a haul truck meeting unexpectedly at a different unlisted intersection. All four were closed within three days with a reminder to crews. None was referred to the plan owner or led to a change.

At 05:10 on 12/08/2026 the Mining Manager directed that a positive call be made before entering any Stage 4 intersection until further notice.

Analysis

External Influences

Guideline sets no review trigger and no basis for the positive-call list

Bindarra's traffic management plan is made under Mines Safety Guideline MSG-14 revision 4. The guideline is issued by the State Mines Inspectorate under regulation 6.22 of the Mining Operations Safety Regulations 2018. Compliance with it is one way of discharging the regulation, which itself requires only that "the operator shall manage the risks arising from the interaction of vehicles at the mine". The guideline's text and its regulatory basis are documentary and not in dispute, so it is virtually certain that the guideline set no review trigger and no basis for nominating positive-call intersections.

Three clauses carry the point.

- Clause 3.2 requires the plan to be reviewed "at intervals not exceeding twelve months, and following any significant change to the mine's road network, vehicle fleet or operating pattern". The guideline does not define a significant change and gives no example of one.
- Clause 5.1 requires the operator to "nominate the intersections, crossings and other interaction zones at the mine at which positive communication between vehicles is required before entry". It sets no minimum, and it does not say who nominates or on what basis.
- Clause 3.5 provides that the plan "is a site document. It is not approved by, and need not be submitted to, the Inspectorate, except on request following an occurrence or an inspection."

Clause 5.4, which allows a light vehicle to be made detectable by any one of five means, is context to this factor and not part of it.

On the day of the occurrence the plan met the guideline on its face. It had been reviewed on 12/11/2025, inside twelve months, and Appendix B nominated nine intersections. The mine was not in breach. This is why the matter is treated as an external influence and not as a compliance finding. The concern is the shape of the discretion the guideline leaves, not any failure by the Inspectorate.

The guideline bears on one path only, which is how an intersection reaches the positive-call list. BIN-TMP-004 clause 9.1 repeats the guideline's wording, requiring review "annually, and following any significant change to the mine's road network". It does not define the term either. The note to Appendix B provides that the appendix "is reviewed and reissued at the annual review of this plan". It also provides that an intersection is added only where it appears on the Schedule 1 road layout drawing, which is reissued at the same review. No provision allows an addition between reviews, and none requires a new road to be referred to the plan's owner.

The document register shows how that operated. No out-of-cycle review had been held in the plan's eleven year history. Of the 14 new roads cut in the Stage 4 pit since the 12/11/2025 review, the 11 cut under a works order were not referred to the plan's owner. The RL 312 spur was one of them. Four hazard reports in the preceding 24 months described a light vehicle and a haul truck meeting unexpectedly at an intersection not listed at Appendix B. Each was closed with a reminder to the crew, and none changed Appendix B, Schedule 1 or any physical control.

The case for influence is that both open terms were left to the same site document. That document was reviewed once a year, and no one outside the mine looked at either term. A defined trigger

would have required the plan to be reopened when a road was added to the network. A stated basis for nomination would have given an intersection that failed the plan's own sight distance rule at clause 6.2 a claim on the list. With either in place, the RL 312 spur and the other unreferred roads would have had a defined route onto the call list between annual reviews.

The case against is substantial. The guideline did not stop anyone acting. Bindarra could have held an out-of-cycle review under its own clause 9.1 at any time, and MSG-14 does not prevent a mine from listing every intersection it has. On this reading, the mine did not fail to act because the guideline was vague. It failed to act because nothing inside the mine put the question in front of the plan's owner. The external clause would then be background, and the organisational shortcomings would account for the whole of the outcome.

The investigation accepted that the mine could have done both things without prompting. What the guideline did was make doing nothing the default. Where a significant change is undefined, and nomination has no stated basis and no outside review, an annual cycle satisfies the guideline whether or not new intersections have been cut in the meantime. On that basis it was assessed as likely that the guideline's open terms contributed. That assessment meets the standard required for a contributing factor and goes no higher. It should not be read as settled beyond that.

The guideline's influence does not extend to the works order. Whether a works order is exempt from management of change is set by the operator's own procedure, ORR-PRO-022 clause 5.3, and not by MSG-14. The link from the works order to the absent controls at the spur is a matter of organisational influence and is argued there. The plan's restriction of positive calls to listed intersections, added only at the annual review, is the control this factor fed. It is argued under risk controls.

It is likely that MSG-14's undefined review trigger and its open basis for nominating positive-call intersections allowed the mine's plan to update its positive-call list only once a year, which left the RL 312 spur intersection off that list on the night of the collision.

Organisational Influences

Spur opened under a works order, so nothing was referred to anyone

ORR-PRO-022 clause 5.2 requires a management of change record before any change to the mine's road network. Clause 5.3 excludes routine earthworks carried out under a works order within an approved pit design. WO-26-4471 relied on that exclusion. It classified the 240 m access to bore DW-14 as "Routine earthworks within the approved Stage 4 design" and recorded management of change as not required. Its handover field read "N/A". The procedure, the works order and the survey department's road register agree with each other. It is therefore virtually certain that the spur was opened without a management of change record.

The consequence followed from where the referrals sit. Clause 5.4 places them inside the management of change record. They go to the traffic management plan owner, to the survey department for the fleet management system road network, and to the contractor's technical services group for the collision awareness system configuration. The works order form at Schedule 4 has no equivalent fields, and none of the three referrals was made. The in-cab map published on 24/07/2026 was still current on the night and showed no intersection at RL 312. The S4-LOAD suppression geofence had been carried up the ramp on 09/06/2026 and already covered the spur mouth.

Had a management of change record been raised, the new intersection would have gone to the plan's owner, who had the means to require the clause 6.2 and 6.4 controls. It would also have gone to the technical services group, who configure the geofence the intersection was cut inside.

Both referrals are standing parts of the procedure, and they are used in practice. Since the 12/11/2025 review, all three roads raised under a management of change record were referred to the plan's owner. None of the eleven raised under a works order was.

A referral is not itself a control. The plan's owner might have left the spur to the annual review, and no out-of-cycle review has been held in the plan's eleven years. A referral also could not have changed the rule that limits positive calls to listed intersections. That rule is argued under Risk Controls.

The opening of the spur under a works order very likely contributed to the absence of a stop control and warning signs at the intersection, and to the suppression of collision alerts there.

Access cards issued with no check of the pit driving assessment

ORR-STD-009 clause 7.3 allows a contractor's own general induction to be accepted where the contractor holds a recognition agreement. Hallorn Survey & Geotechnical has held one since 2023. Clause 7.4 states that this equivalence "does not extend to the Bindarra site-specific supplement, or to the Bindarra light vehicle pit driving assessment". Clause 7.5 makes the engaging department responsible for verifying 7.4 before a card is issued.

The standard is clear. What it lacked was a place for the verification to happen. The gatehouse issues cards against the recognition agreement alone. Nothing in the access card system distinguishes a person who may drive in an operating pit from one who may not. The light vehicle driver's card was issued on 09/08/2026 after a 20 minute briefing. His training file showed the Hallorn online module of 24/07/2026 and no pit driving assessment. It is virtually certain that the card was issued with no check of the assessment. Whether he was given the site supplement is [NOT ESTABLISHED: Hallorn records of the issue of the Bindarra site-specific supplement].

The case for an influence rests on what the assessment teaches. It covers giving way to haul trucks, approaching an intersection a driver cannot see up, and when calls are made. Without a completed assessment he would not have been carded to drive in the pit. The driver stated that he would have called if he had been told to.

The case against is substantial. He was already applying the assessment's first rule, that light vehicles give way to heavy vehicles at all times. He slowed and looked before he went. The plan requires no call at an unlisted intersection, so the assessment would not have taught him to call at RL 312. His statement about calling cannot be extended that far.

What the assessment adds over what he did is narrower. At a mouth he cannot see out of, he should stop and look again rather than slow. On the recorded times, a stop at the mouth would very likely have shown him the truck. No copy of the assessment's content was available, and that limits the strength of this conclusion.

The issue of access cards with no check of the pit driving assessment likely contributed to the light vehicle driving out of the spur onto the ramp without stopping.

Haul operators rostered for fourteen consecutive 12.5 hour night shifts

The contractor's Stage 4 haul roster is 14 nights on and 7 off, from 18:00 to 06:30 local. The roster, the site access records and the fatigue standard are documentary, so the roster is virtually certain. The haul truck operator was on night 10 of 14. The collision at 02:47 fell in the ninth hour of his shift, inside the window of lowest circadian alertness. Every limit in the standard was met.

The mechanism a roster like this works through is degraded detection and slowed response. The recorded data contradict that mechanism here. The operator's response was 1.4 seconds, inside the 1.5 seconds the plan's own stopping distance calculation assumes. The braking and steering that followed were correct and immediate. At the recorded 0.78 m/s² from 53 m, a response time

of zero would still have ended in a collision, at about 17 km/h rather than 24 km/h. The download shows no decision or omission for the roster to have acted through. It is therefore unlikely that the roster influenced this occurrence. The operator's own state on the night is a separate question, discussed under Local Conditions.

The roster remains a standing exposure. On every swing it puts each haul truck operator at the wheel of a 220 tonne vehicle in the small hours of a tenth, twelfth and fourteenth consecutive night. At those times, the defence against an unlit hazard at an unlisted intersection is the operator's own detection and response. At an intersection with a longer sight line than RL 312, a slower response could decide whether the truck stops in time.

The roster of fourteen consecutive 12.5 hour night shifts likely increased risk to haul operations.

Risk Controls

Collision alerts suppressed at the spur by a geofence operators knew nothing of

The S4-LOAD geofence was drawn in 2025 to stop nuisance alerts at the loading face, and the investigation did not question that purpose. What mattered was its extension on 09/06/2026. That change carried the boundary up the main ramp for the first time, to a point 180 m above where the RL 312 spur was cut about seven weeks later. The contractor's technical services group made the change as a configuration task. It was not recorded in the traffic management plan or reviewed against it. No training record on site covered the geofences. The configuration, the event log and the fleet statistics came from the system itself. The operators' lack of knowledge rested on the accounts of the truck operator and the night shift supervisor and on the absence of any training record or notice. On that basis it is extremely likely that alerts were suppressed at the spur by a geofence that operators did not know about.

The alert itself would have gained nothing. Unsuppressed, it would have sounded at about 02:47:06.5, half a second after the beacon came into view and about a second before the brakes were applied. It would have told the operator what he was already reacting to, and the system is advisory only. The influence lay in expectation instead. Before June the ramp was outside the geofence, and a quiet system there meant an empty ramp. From June the system was silent there whatever was on the road. The operator wrote in his own report, before he was asked, that he had taken a quiet pit to mean an empty one. He came over the crest with that expectation, on a ramp that showed no intersection on his screen or on the road.

Against this, the expectation was not shown to have changed what the operator did. He was descending at the speed every operator used on that ramp, and his own remedy was a sign rather than the system. The effect is real but cannot be measured, and both readings stand. It is likely that the suppressed system contributed by reinforcing the operator's belief that the ramp was empty.

Positive calls required only at listed intersections, added only at the annual review

The mine's defence against this interaction was the positive call. Under clause 4.3.2 it was not required at the RL 312 spur. An intersection reached Appendix B only through Schedule 1. Schedule 1 was reissued only at the annual review, and nothing required a new road to be referred to the plan's owner. The spur was on neither document. It is virtually certain that the plan, as written, could not have required a call at this intersection before 12/11/2026.

The pit channel recording shows that the control worked where it applied. HT-407 called at the RL 360 crossover going up and coming down and at the ROM pad turn-off. Each call was acknowledged, and the last was made 51 seconds before the collision. No call was made at the spur. The truck operator, the light vehicle driver and the night shift supervisor each described the same practice independently. Had the spur been listed, the call would have been made on a

channel the light vehicle was on. A call made but not acknowledged would still have left a driver to decide whether to go. It is very likely that restricting calls to listed intersections contributed to neither driver making one. The absence of any review trigger in the guideline behind the list is discussed under External Influences. The drivers' actions are discussed under Individual Actions.

No stop control or warning sign at an intersection failing the sight distance rule

Clause 6.2 applied at every intersection, listed or not. The survey gave 41 m of sight distance against a 96 m stopping distance, so the clause required a stop control, signalised control or traffic controller. Clause 6.4 signage applied only to intersections on Schedule 1. The intersection had no warning sign, no stop or give-way control, no delineation and no posted speed. The construction sign had been removed on 02/08/2026 and nothing replaced it. HAZ-2026-0917 named both omissions seven days before the occurrence. It is virtually certain that the intersection lacked the controls the plan required.

The case for influence rests on timing. The truck was within the light vehicle driver's 38 m of view from about 02:47:07.4, nearly two seconds before the light vehicle reached the mouth. A full stop there would very likely have shown him the truck, and he could then have let it pass, as the technician did on 04/08/2026.

The contrary evidence was weighed. A 38 m view gives about four seconds of a truck at 37 km/h, which is a thin margin even from a standstill. The driver was already slowing and looking under a give-way rule. The effect of a sign on the truck rests on the operator's account after the event. A sign at an intersection absent from the in-cab map would also have been a single cue. For these reasons the finding sits below the others. It is likely that the absence of a stop control and warning signs contributed to the light vehicle entering the ramp without stopping.

Night shift extrication waits on a second responder called from the village

Night shift rosters one mines rescue paramedic. The second responder must be present before any extrication begins. That person is called from the village 11 km from the gatehouse, in the vehicle that carries the hydraulic rescue tool. The paramedic's primary survey was complete at 02:58. The second responder arrived at 03:07, and extrication followed at 03:14. The post-incident review measured the delay at approximately 9 minutes. It is virtually certain that this arrangement existed.

The collision and the injuries were complete before the team was paged. The review found that the delay did not affect the clinical outcome, because the driver was conscious and stable throughout. It is unlikely that the arrangement affected this occurrence or its outcome.

The arrangement remains important. Night shift extrication capability is one person plus a drive from the village, and nothing in it scales with severity. With a fuel fire, an unconscious casualty or a compromised airway, the same nine minutes could change the outcome. It is very likely that the arrangement increased risk.

Local Conditions

Sight line of 41 m at the spur against a 96 m stopping distance

The mine survey department measured the intersection from the eye heights of both vehicles. From the truck cab, the beacon on a light vehicle in the spur first came into view at 53 m. The vehicle itself came into view at 41 m. From the spur mouth, a driver first had sight of a descending truck at 38 m, and the inner windrow hid the truck's lower body for the last 25 m of the spur's approach. A 220 t class truck travelling empty at the posted 40 km/h on this grade needed 96 m to stop. The technician's hazard report of 04/08/2026 had described the same geometry from the same

position: "you cannot see up the ramp from that road". It is virtually certain that the sight line existed as surveyed.

The recorded approach shows what that geometry meant. HT-407 was travelling at 37 km/h when the beacon first became visible. From that speed, at the 0.78 m/s² the truck actually achieved, it needed about 68 m on brakes alone. That figure makes no allowance for any response by the operator. Neither 53 m nor 41 m came close to it. With twice the sight line, the same 1.4 second response and the same braking would have stopped the truck short of the spur. The light vehicle driver was no better placed, because he could not see a descending truck until he was at the mouth. No degree of attention from either driver could have made this intersection safe.

The missing stop control and warning signs are a defence rather than geometry, and they are discussed under Risk Controls.

The 41 m sight line at the spur, against a 96 m stopping distance, very likely contributed to neither driver being aware of the other until the truck came over the crest.

Airborne dust on the ramp above the intersection during the truck's approach

The first question was whether dust hung on the ramp above the crest. Several points supported it. The water cart's last pass on that section had been at 01:10, 1 hour 37 minutes before the collision, so a pass was 7 minutes overdue. The fixed monitor on the RL 328 bench stands above the grade break the truck came over. In the ten minutes covering the occurrence it logged PM10 at 142 micrograms per cubic metre, against a shift mean of 88. The truck operator described dust hanging on the descent, which he made behind his own lights into an unlit pit.

The light vehicle driver was firm that the air was clear. His account was accepted as honest. However, it described the spur below the windrow, where he was moving slowly or stopped, and not the ramp above the crest where the truck was. The reading was also well below the site trigger of 250. It measured particulate at the monitor head and not what could be seen from either cab, and nothing at Bindarra measures visibility. On balance, dust was likely present on the ramp above the intersection.

The second question was whether that dust delayed the operator's sighting of the beacon. One argument was that dust in a truck's lights over a crest is the condition in which a driver picks out a beacon late. The argument against was stronger. The survey puts the first possible sight of the beacon at 53 m, even in clear air. The download puts the truck at that point at its last steady data point. A sighting at the geometric limit leaves dust nothing to explain. The operator himself would not say the dust made a difference. His 1.4 second response was inside the 1.5 seconds the plan's stopping distance assumes, which leaves little room for a delay.

The matter could not be closed, because the moment of sighting was inferred from the data and the survey and was not recorded. Those sources cannot exclude dust costing a fraction of a second at the edge of the truck's lights. Settling the question would need a measurement of visibility from the cab, or a direct record of when the operator first saw the beacon. Neither exists.

Whether the dust delayed the sighting remains unresolved at about as likely as not, and nothing from it is carried into the findings.

Ramp surface wet or slippery at and above the intersection

It is very unlikely that the ramp was wet or slippery. No rain had fallen for 26 days and the ramp was found dry at 06:20. The braked scuff ran 58 m with no sign of a sliding wheel, and the truck achieved 0.78 m/s² against its own tested dry figure. The surface was therefore excluded.

Haul truck operator fatigued during the approach to the intersection

The operator was on night 10 of a 14 night roster, 8 hours 47 minutes into his shift, at 02:47. This is within the period of lowest circadian alertness. His voluntary sleep record showed between 5 and 7 hours on each of the ten days before, with 6 hours 30 minutes on the last of them. He described himself as tired in the way usual for a tenth night. The task on the approach was detection in an unlit pit, which fatigue degrades first.

Against this, he described the tiredness as ordinary and said he was awake and driving properly. Every limit in the contractor's fatigue standard was met. His recorded performance showed no sign of impairment. He responded within 1.4 seconds, and his braking and steering were correct and immediate. Feeling tired is not the same as being fatigued to a degree that degrades performance, and no other evidence bore on the question. A detailed fatigue assessment would have been needed to settle it.

The outcome would not have changed either way. Braking at 0.78 m/s² from 53 m with no response delay at all still ends in a collision, at about 17 km/h rather than 24 km/h.

Whether the operator was fatigued remains unresolved at about as likely as not, and nothing from it is carried into the findings. The roster is considered separately under Organisational Influences.

Individual Actions

Neither driver made a positive call before entering the intersection

The pit channel was logged continuously from the return from crib at 02:22:41 until after the collision. No transmission by either vehicle referred to the RL 312 spur, the dewatering bore or a light vehicle in the Stage 4 pit. LV-118 made no transmission that shift after logging on at the start of shift. All three interviews confirmed this. It is virtually certain that neither driver made a call.

The omission matched what the plan required. Clause 4.3.2 of BIN-TMP-004 requires no call at an intersection not listed at Appendix B, and the RL 312 spur was not listed. The truck operator called at every listed intersection he passed, and each call was acknowledged. His last, "407, three-sixty crossover, empty, going down through.", came 51 seconds before the collision.

The truck operator, the light vehicle driver and the night shift supervisor each described the same practice independently. Calls were made at the nine intersections on the crib room board and nowhere else. The supervisor had never corrected anyone for not calling elsewhere. Neither driver departed from the practice of the pit. Why the practice took that form is argued under Risk Controls, where the limit of positive calls to listed intersections is found to have contributed.

A call from either vehicle would have been heard by the other, by dispatch and by the supervisor. The dispatcher was already tracking the truck closely enough to place it, and at 02:46:18 told the excavator operator it was about four minutes off. A call would therefore have put each vehicle in the other's knowledge before either reached the intersection. A call gives knowledge, though, not separation. The two vehicles would still have had to settle which went first across 41 m of sight line.

It is likely that the absence of a positive call from either vehicle contributed to neither driver being aware of the other until the truck came over the crest.

Light vehicle drove out of the spur onto the ramp without stopping

The driver's account is that he slowed at the mouth, looked right up the ramp, saw nothing and drove out, turning right to go up the ramp. On read-back he corrected his account from stopped to slowed. He put his speed leaving the spur at about 25 km/h. That correction ran against his own interest. His account agrees with the collision awareness log, which placed LV-118 at the spur mouth at 02:47:09.2 with the truck 21.4 m above it. It also agrees with the impact position, 6.2 m

into the ramp 2.8 seconds later. It is virtually certain that the light vehicle drove onto the ramp without stopping.

The timing explains why the look found nothing. From the mouth his sight line to a descending truck was 38 m. The windrow hid the truck's lower body over the last 25 m of the spur. On the recorded times his look came a second or two before the truck came over the crest, when it was beyond anything his position let him see. The truck was inside his 38 m from about 02:47:07.4, nearly two seconds before he reached the mouth.

He applied the rule he understood. Clause 4.1 requires light vehicles to give way to heavy vehicles at all times. Nothing at the mouth told him the intersection needed more. There was no stop or give-way control, no stop line, no warning sign and no posted speed. He had not done the Bindarra pit driving assessment and had never seen the traffic management plan. The missing stop control is argued under Risk Controls. The issue of his access card without the assessment is argued under Organisational Influences.

Driving out without stopping was the step that put the light vehicle across the truck's path. Had he stopped and looked again, the truck was already within view from the mouth. He would very likely have seen it and let it pass, as the dewatering technician did at the same mouth on 04/08/2026 by reversing. The qualification is that a stop is only as good as the look from it, and that look reached 38 m.

It is extremely likely that the light vehicle driving onto the ramp without stopping contributed to neither driver being aware of the other until the truck came over the crest.

Truck operator braked and steered right within 1.4 seconds of the beacon appearing

The download recorded the last steady data point at 02:47:06.0, at 37 km/h and 53 m above the spur mouth. That is where the survey places the first sight of a light vehicle's beacon over the crest. Service brake pressure rose from 0 to 620 kPa in 0.4 seconds from 02:47:07.4. The retarder went to maximum and 14 degrees of right steering was applied in the same movement. The braked scuff began 38.5 m above the mouth and deviated 1.4 m right over its first 20 m, which matched the download. It is virtually certain that the operator braked and steered right within 1.4 seconds of the beacon appearing.

The response was inside the 1.5 seconds that the plan's own stopping distance calculation allows. The truck achieved 0.78 m/s² net, against the unit's tested 0.80 m/s². It struck at 24 km/h, six seconds after the beacon came into view. Without braking it would have struck at 37 km/h. At 24 km/h the impact energy was about 0.42 of its value at 37 km/h.

The steering mattered as much as the braking. To the operator's left lay the pit edge and a drop of more than 100 m. Steering towards the wall kept the push running down the ramp. LV-118 came to rest against the pit side windrow rather than being driven through it. The truck stopped 19 m below the intersection. The operator called the collision and stayed in the cab as dispatch instructed. He then stayed with the driver until the response team arrived.

It is very likely that the operator's braking and steering reduced the severity of the collision and kept the light vehicle on the ramp.

Occurrence Events

Neither driver aware of the other until the truck came over the crest

Every line of enquiry in this investigation arrived at one condition. Two vehicles were converging on the same place at night, and neither driver knew of the other until six seconds remained.

HT-407 was descending empty at 37 km/h. Its in-cab road network, last published on 24/07/2026, showed no intersection at RL 312. There was no warning sign on the approach and no call on the pit channel. The collision awareness system logged the light vehicle at 21.4 m and raised nothing in the cab. LV-118 was approaching a mouth with 38 m of sight line up the ramp, no give-way control and no call. The night shift supervisor did not know a light vehicle was in the pit. There was no permit, call-in or board by which he could have known.

The survey placed the first sight of a light vehicle's beacon from the truck's cab at 53 m. The download put the truck 53 m above the spur mouth at 02:47:06, its last steady data point and six seconds before the impact. The two agree. The light vehicle driver looked up the ramp before the truck had come within his view. The pit channel recording, the download, the collision awareness system log and the three interviews gave no indication that either driver knew of the other before that moment.

The conditions behind this convergence are argued in their own subsections:

- the 41 m sight line against a 96 m stopping distance
- the suppression geofence
- the absence of a positive call
- the light vehicle entering the ramp without stopping.

Whether airborne dust affected the moment of first sighting could not be resolved, and that matter is discussed under Local Conditions. The conditions are treated here as one mechanism because they acted together on a single state of mutual unawareness.

Three independent recordings and all three interviews established the convergence. The one part not independently corroborated was the light vehicle's movement before it reached the spur mouth, which rested on the driver's account alone. Once the vehicles were six seconds apart and each unaware of the other, nothing between that point and the impact could have prevented the collision.

It was extremely likely that neither driver was aware of the other until the truck came over the crest, and extremely likely that this led directly to the collision.

The haul truck's front left struck the right side of the light vehicle at 24 km/h

The truck struck LV-118 at 02:47:12 at 24 km/h. The brakes had been applied 4.6 seconds earlier. The impact was 6.2 m into the ramp from the spur mouth and 3.1 m out from the toe of the inner windrow. The truck's accelerometer logged 1.9 g longitudinal. Damage to LV-118 was concentrated on the right side from the B pillar forward. Damage to HT-407 was confined to the front left bumper, the left mudguard and the lower two steps of the access ladder, with nothing above the bumper line. Right-side glass and mirror fragments lay from 4.8 m to 14 m down-ramp of the impact. The recorded data, the damage pattern and the debris field all agree.

The intersection geometry dictated the side impact. The spur joined the ramp on a descending truck's left, the open side, on the inside of a left-hand bend at an included angle of 71 degrees. A light vehicle turning right to go up the ramp therefore had to cross the descending lane with its right side towards any truck coming down. At this intersection a meeting between the two could only be side-on.

When a 220 tonne vehicle strikes the side of a 3 tonne one, the outcome depends on the light vehicle's structure, the restraint and the speed. Structure and restraint were settled before the sighting. Speed was the only variable left within the six seconds. The braking brought the impact speed down from the 37 km/h it would otherwise have been to 24 km/h. The operator's response is argued under Individual Actions.

It was virtually certain that the truck's front left struck the right side of the light vehicle at 24 km/h, and virtually certain that this impact was how the convergence became an injury.

The light vehicle was pushed 18 m down the ramp and rolled onto its left side

A continuous gouge ran 18.4 m down-ramp from the impact. At 11.2 m it changed from tyre scuff to body panel gouge, which marks where the vehicle went over. LV-118 came to rest on its left side against the inner windrow, facing down-ramp. The driver sustained a fractured left clavicle, three fractured left ribs and a small pneumothorax, and was conscious throughout. The scene measurements, the vehicle examination and the medical evidence agree.

The roll-over was the means by which the collision produced these injuries. Two protections worked. The roll-over protection structure was deformed inward at the right rear mounting but was not breached. The driver's seatbelt was found fastened, and its webbing showed loading marks. The injuries were those a side impact and roll-over produce when the structure holds and the restraint is worn. Without either protection, the same sequence would have had a different outcome.

The door on the resting side was against the ground, so the driver was extricated through the rear of the vehicle. The time taken to extricate him is considered under Risk Controls.

It was virtually certain that the light vehicle was pushed down the ramp and rolled onto its left side, and virtually certain that this produced the driver's injuries.

Service brakes delivered less than their tested deceleration during the stop

This hypothesis was ruled out as very unlikely. The 0.78 m/s² net achieved during the stop was within the normal scatter of the 0.80 and 0.82 m/s² net measured in tests on 03/08/2026 and 14/08/2026, and no brake, retarder or park brake defect had been raised against the unit in the preceding twelve months.

Conclusions

These findings are made with respect to the occurrence and should not be read as apportioning blame or liability to any organisation or individual.

A finding stated without a likelihood term was established as fact.

Findings

1. It was virtually certain that Mines Safety Guideline MSG-14 revision 4 did not define a significant change that would trigger a review of the traffic management plan, and set no minimum and no basis for nominating positive-call intersections. It was likely that this contributed, by leaving the listing of a new intersection to an annual site review with no defined route onto the list between reviews. That finding was held at the standard of proof and no higher, because the mine could have reviewed its plan or listed any intersection on its own initiative and was not in breach of the guideline. (Safety issue)

2. It was virtually certain that the RL 312 spur was cut under works order WO-26-4471 as routine earthworks excluded from management of change by ORR-PRO-022 clause 5.3. As a result, none of the three referrals that clause 5.4 would have triggered was made, whether to the traffic management plan owner, the survey department or the contractor's technical services group. It was very likely that this contributed by leaving the new intersection without the controls, in-cab road network update and geofence review that a referral would have reached. (Safety issue)

3. It was virtually certain that the light vehicle driver was issued a site access card against the contractor's recognition agreement alone. The access card system held no field for the Bindarra light vehicle pit driving assessment that ORR-STD-009 clause 7.4 required. It was likely that this contributed to the light vehicle driving out of the spur without stopping, because the assessment teaches a driver to stop and look again at a mouth he cannot see out of. That finding was held below others because the content of the assessment was not available, and because the driver was already applying its first rule. (Safety issue)

4. It was virtually certain that the Stage 4 haul roster placed operators on up to fourteen consecutive 12 hour 30 minute night shifts, within the contractor's fatigue standard. It was likely that this increased risk across the haul fleet. It did not contribute to this occurrence, because the operator's recorded 1.4 second response was within the plan's own allowance, and a response time of zero would still have ended in a collision. (Safety issue)

5. It was very likely that the S4-LOAD suppression geofence covered the place where the spur joined the main ramp, and that operators did not know collision alerts were suppressed there. The geofence had been extended up the ramp on 09/06/2026 without review against the traffic management plan. It was likely that this contributed, through the operator's expectation that a silent system meant an empty ramp, rather than through the alert itself. An unsuppressed alert would have sounded about half a second after the beacon came into view, and the effect of that expectation could not be measured. (Safety issue)

6. It was virtually certain that BIN-TMP-004 required positive calls only at the nine intersections listed at Appendix B. An intersection could be added to that list only at the annual review, through a road layout drawing reissued at that review, and no provision required a new road to be referred to the plan's owner. It was very likely that this contributed, because the RL 312 intersection was not listed and the truck operator called at every listed intersection he passed that night. (Safety issue)

7. It was virtually certain that there was no stop or give-way control, warning sign, delineation or posted speed at the RL 312 intersection, although its 41 m sight distance failed the clause 6.2 rule.

It was likely that this contributed to the light vehicle driving out without stopping. That finding was held below others because the margin from a stop at the mouth was thin, and because the effect a sign would have had on the truck rested on the operator's own account. (Safety issue)

8. It was virtually certain that the night shift extrication capability depended on a second responder called from the accommodation village 11 km from the gatehouse, in the vehicle carrying the hydraulic rescue tool. The post-incident review measured the resulting wait at approximately 9 minutes. It was very likely that this increased risk. It did not contribute to this occurrence or its outcome, because the injuries were complete before the response was paged and the driver was stable and conscious throughout. (Safety issue)

9. It was virtually certain that the available sight distance at the RL 312 intersection was 41 m against a 96 m stopping distance for an empty 220 t class truck at the posted 40 km/h. It was very likely that this contributed, because with the same response and braking, twice the sight line would have stopped the truck short of the spur.

10. The ramp surface at and above the intersection was examined and was not wet or slippery. That hypothesis was ruled out.

11. It was virtually certain that neither driver made a positive radio call before entering the intersection, where the traffic management plan required none. It was likely that this contributed, because a call would have been heard by the other vehicle, by dispatch and by the supervisor. That finding was qualified because a call gives knowledge and not separation, and the two vehicles would still have had 41 m of sight line in which to resolve who went first.

12. It was virtually certain that the light vehicle slowed at the spur mouth and drove out onto the ramp without stopping. On the recorded times, the driver's look up the ramp came before the truck was within his 38 m of view, and the intersection had no stop control or sign. It was very likely that this contributed, because a stop at the mouth would very likely have shown him the truck.

13. The truck operator braked, applied the retarder and steered 14 degrees right within 1.4 seconds of the beacon coming into view. This reduced the impact speed from 37 km/h to 24 km/h and turned the truck away from the open pit edge, so the light vehicle came to rest against the inner windrow.

14. It was very likely that neither driver was aware of the other until the truck came over the crest, six seconds before the impact. It was very likely that this contributed to the collision that followed.

15. It was virtually certain that the haul truck's front left struck the right side of the light vehicle at 24 km/h, and virtually certain that this contributed to the injuries.

16. It was virtually certain that the light vehicle was pushed 18.4 m down the ramp and rolled onto its left side against the inner windrow, and virtually certain that this produced the driver's injuries. The seatbelt was fastened and loaded, and the roll-over protection structure deformed without being breached.

17. The service brakes were examined and delivered their tested deceleration during the stop. That hypothesis was ruled out.

Probable cause

The RL 312 spur was opened under a works order that excluded it from management of change, so a new intersection with 41 m of sight distance against a 96 m stopping distance went into service unreviewed to the traffic management plan's owner and absent from the positive-call list, with the result that neither driver was aware of the other until the truck came over the crest, the light vehicle drove out onto the ramp without stopping, and the descending haul truck struck its right side.

Contributing factors

- It was virtually certain that the spur was opened under a works order, so nothing was referred to anyone, and very likely that this contributed to the occurrence. (Safety issue)
- Collision alerts were very likely suppressed at the spur by a geofence operators knew nothing of, and this likely contributed through the operator's expectation that a quiet system meant an empty ramp. (Safety issue)
- The sight line of 41 m at the spur against a 96 m stopping distance was virtually certain, and it very likely contributed.
- With virtual certainty, the guideline set no review trigger and no basis for the positive-call list, and it likely contributed, at the standard of proof and no higher. (Safety issue)
- Positive calls were, with virtual certainty, required only at listed intersections, added only at the annual review, and this very likely contributed. (Safety issue)
- That neither driver made a positive call before entering the intersection was virtually certain, and the omission likely contributed, although a call would have given knowledge rather than separation.
- It was virtually certain that access cards were issued with no check of the pit driving assessment, and likely that this contributed. (Safety issue)
- It was virtually certain that there was no stop control or warning sign at an intersection failing the sight distance rule, and likely that the absence contributed. (Safety issue)
- The light vehicle virtually certainly drove out of the spur onto the ramp without stopping, which very likely contributed.
- Neither driver was very likely aware of the other until the truck came over the crest, and that condition very likely contributed.
- With virtual certainty, the haul truck's front left struck the right side of the light vehicle at 24 km/h, and the impact contributed with virtual certainty.
- The light vehicle was, with virtual certainty, pushed 18 m down the ramp and rolled onto its left side, and this contributed with virtual certainty.

Other factors that increased risk

- It was virtually certain that haul operators were rostered for fourteen consecutive 12.5 hour night shifts, and likely that this increased risk, although it did not contribute to this occurrence. (Safety issue)
- The night shift extrication virtually certainly waited on a second responder called from the village, which very likely increased risk, although it did not contribute to this occurrence or its outcome. (Safety issue)

Other findings

- The hypothesis that the ramp surface was wet or slippery at and above the intersection was examined and ruled out. No rain had fallen for 26 days, the investigator found the surface dry at 06:20, and the truck achieved its dry-grade deceleration.
- The hypothesis that the service brakes delivered less than their tested deceleration during the stop was examined and ruled out. The occurrence download gave 0.78 m/s² net, against tests of 0.80 before the occurrence and 0.82 after it, and no defect history existed.
- The truck operator braked and steered right within 1.4 seconds of the beacon appearing, which was found to reduce risk.

- Whether airborne dust on the ramp above the intersection delayed the truck operator's sighting of the beacon could not be determined either way. Dust was likely present, with a water cart pass 7 minutes overdue and a PM10 reading of 142 micrograms per cubic metre against a shift mean of 88. Against that, the beacon was in view from 53 m, the first point at which the geometry allowed it to be seen at all, and the operator's response from that point was within the plan's allowance. The sighting point was inferred from the last steady data point and the survey rather than recorded, and nothing at Bindarra measures visibility, so a delay of a fraction of a second could not be excluded.
- Whether the haul truck operator was fatigued during the approach to the intersection could not be determined either way. His duty period, the hour, his recorded sleep and his own description of being tired pointed towards fatigue. His account of being awake and driving properly, and his recorded 1.4 second response with correct braking and steering, pointed against impairment. A detailed fatigue assessment would be needed to settle it. Even if fatigue had been present, a response time of zero would still have ended in a collision, at about 17 km/h instead of 24 km/h.

Recommendations

Guideline sets no review trigger and no basis for the positive-call list

Mines Safety Guideline MSG-14 revision 4 requires a traffic management plan to be reviewed after any significant change to the road network. It does not define a significant change. It also requires the operator to nominate the intersections at which positive communication is required, but sets no minimum and no basis for that nomination. The plan is a site document. It is not approved by the Inspectorate and is not submitted to it except on request after an occurrence. On the day of the occurrence, Bindarra's plan met the guideline on its face, and the mine was not in breach.

The issue lies in the width of the discretion the guideline leaves, not in any failure by the Inspectorate. Neither term is defined. As a result, both the decision to reopen the plan and the choice of positive-call intersections were left to the same site document, reviewed once a year with no outside view of either. It is likely that this contributed to positive calls being required only at listed intersections, with new intersections added only at the annual review.

That finding is held on the balance and no higher. The mine could have held an out-of-cycle review under its own clause 9.1 at any time. Nothing in the guideline prevented it from listing every intersection it had. What the guideline did was make inaction the default.

No safety action has been recorded against this issue, and no recommendation is made.

Spur opened under a works order, so nothing was referred to anyone

The mine's management of change procedure, ORR-PRO-022, excluded routine earthworks carried out under a works order within an approved pit design. Works order WO-26-4471 relied on that exclusion when it cut 240 m of access to bore DW-14 at RL 312. That work created a new intersection with the main ramp, but no management of change record was raised. As a result, none of the three referrals the procedure holds at clause 5.4 was made. The traffic management plan owner, the survey department and the contractor's technical services group were not told of the new intersection. This very likely contributed to the intersection having no stop control or warning sign. It also contributed to the collision alerts at the spur remaining suppressed by the S4-LOAD geofence.

The issue extends beyond this intersection. Fourteen new roads have been cut in Stage 4 since the 12/11/2025 plan review. The three raised under a management of change record were referred to the plan's owner. None of the eleven raised under a works order was.

Recommendation

The investigation recommends that [NOT ESTABLISHED: the organisation, by role, to which this recommendation is addressed] amend ORR-PRO-022 so that earthworks creating a new intersection with a haul road are no longer excluded under clause 5.3. Alternatively, the works order form at Schedule 4 should carry the same three referrals that clause 5.4 attaches to a management of change record:

- the traffic management plan owner
- the survey department, for the fleet management system road network
- the contractor's technical services group, for the collision awareness system geofences

A response was due by 30/09/2026. [NOT ESTABLISHED: response to the recommendation and any safety action taken against this issue]

Access cards issued with no check of the pit driving assessment

The Bindarra gatehouse issued access cards against a contractor's recognition agreement alone. Contractor Management Standard ORR-STD-009 clause 7.4 requires every person who will drive in an operating pit to complete the Bindarra light vehicle pit driving assessment at site. Clause 7.5 makes the engaging department responsible for verifying this before a card is issued. The access card system had no field for that verification. Nothing in it distinguished a person who may drive in an operating pit from one who may not. The light vehicle driver was issued his card on 09/08/2026 after a 20 minute gatehouse briefing. He had not completed the assessment.

The standard itself was sound. The safety issue was that there was no point at which its required verification could take place. As a result, a contractor's employee could be issued a card to drive in an operating pit without first completing the site assessment. That assessment covers giving way to haul trucks, approaching an intersection the driver cannot see up, and when calls are made. In this occurrence, issuing the card without the assessment likely contributed to the light vehicle driving out of the spur onto the ramp without stopping.

No safety action has been recorded against this safety issue.

Positive calls required only at listed intersections, added only at the annual review

BIN-TMP-004 required a positive call on the pit channel only at the nine intersections listed at Appendix B. An intersection could join that list only through the Schedule 1 road layout drawing. That drawing was reissued only at the annual review, and nothing required a new road to be referred to the plan's owner. The RL 312 spur was on neither document. This very likely contributed to neither driver making a positive call before entering the intersection.

The issue reaches beyond this intersection. Eleven of the fourteen new Stage 4 roads cut since the last review were never referred. No out-of-cycle review had been held in the plan's eleven year history. By its own terms, the mine's defence against this interaction could not be applied at a new intersection until the next annual review. For the spur, that was 12/11/2026, three months after the collision.

At 05:10 on 12/08/2026, two hours twenty-three minutes after the collision, the Mining Manager directed that a positive call be made before entering any intersection in the Stage 4 pit, listed or not. The direction was read at the following four pre-starts and posted beside the Appendix B board in the Stage 4 crib room. It stands until BIN-TMP-004 is amended. The direction addresses the issue in part. It covers the Stage 4 pit only, and it leaves the listing route and the absence of a referral requirement unchanged.

The investigation recommends that the operator amend BIN-TMP-004 to provide a route by which an intersection reaches Appendix B between annual reviews. The plan should also require a new road to be referred to its owner when it is opened to traffic.

Collision alerts suppressed at the spur by a geofence operators knew nothing of

The collision awareness system on the Stage 4 haul fleet raised no alert inside the S4-LOAD suppression geofence. On 09/06/2026 the contractor's technical services group extended that boundary up the main ramp, to a point 180 m up-ramp of the RL 312 intersection, as a system configuration task. The change was not reviewed against the traffic management plan, and nobody outside that group was told. The spur, cut from 28/07/2026, therefore joined the ramp inside the suppressed area. No training on site covered the geofences.

The issue matters because of what the silence taught operators. An unsuppressed alert would have sounded only as the truck operator was already reacting. Operators had, however, come to read a quiet system on the ramp as an empty ramp, and from June that reading no longer held. That expectation likely contributed to the occurrence. In the twelve months before the occurrence, 68 proximity events were suppressed inside the geofence across the fleet. Nine of them involved a light vehicle on the main ramp below the boundary.

On 28/08/2026, Tarrowen technical services added suppressed proximity events to the monthly collision awareness report, broken down by geofence and by vehicle class. It also re-cut the previous twelve months for the Stage 4 fleet on the same basis. This makes suppressed events visible. It addresses the issue in part, because it neither brings boundary changes under change control nor tells operators where the system is silent.

The investigation recommends that [NOT ESTABLISHED: organisation to which the recommendation is addressed] take two actions. It should treat a change to a suppression geofence boundary as a change to a documented control under ORR-PRO-022 clause 5.2. It should also tell operators in writing where the collision awareness system is suppressed and why.

No stop control or warning sign at an intersection failing the sight distance rule

The RL 312 intersection was in use with none of the controls the traffic management plan required. Its available sight distance of 41 m was less than the 96 m stopping distance, so clause 6.2 called for a stop control, signalised control or traffic controller. None was installed. There was no warning sign on either ramp approach, no give-way control and no delineation at the spur mouth. HAZ-2026-0917 reported the missing sign and give-way seven days before the occurrence.

The absence of these controls likely contributed to the light vehicle driving out of the spur onto the ramp without stopping. A full stop at the mouth would very likely have brought the truck into the driver's view before he entered. The margin from that stop was thin, at 38 m, or about four seconds of a truck at 37 km/h. The controls were missing because the spur was opened under a works order, and nothing was referred to the plan's owner.

The mine operator barricaded and closed the RL 312 spur on the night of the occurrence, and it remains closed. Survey then measured the available sight distance at the other ten unlisted roads cut in Stage 4. Two of the ten were below the clause 6.2 minimum and have been closed pending a stop control, and these remain open actions. This addresses the issue in part. It removes the exposure at the roads identified. It does not by itself ensure that a road opened in future receives the controls clause 6.2 requires, because that depends on the referral addressed under the works order issue. No further recommendation has been made against this issue.

Haul operators rostered for fourteen consecutive 12.5 hour night shifts

The contractor's Stage 4 haul roster is 14 nights on and 7 off, from 18:00 to 06:30 local. Each shift runs 12 hours 30 minutes with a 30 minute crib. The roster sits inside a fatigue management standard that permits up to 14 consecutive night shifts with a minimum 10 hour break. It is virtually certain that this roster was in place. The night shift supervisor works the same pattern as the crews he supervises.

The haul truck operator was on night 10 of 14, and every limit in the standard had been met. It is unlikely that the roster influenced this collision. The operator's recorded response to the beacon fell inside the time the stopping distance calculation assumes. The sight line at the spur left no margin that a faster response could have recovered.

The roster is likely to be a matter of safety significance in its own right. It is a standing exposure built into the roster rather than a feature of one night. On every swing, operators of 220 tonne haul trucks drive in the small hours of a tenth, twelfth and fourteenth consecutive night. At an unlisted intersection with an unlit hazard, the defence is the operator's own detection and response. Elsewhere on the pit, the sight line may be long enough for response time to decide the outcome. No safety action has been recorded against this issue, and no recommendation is made.

Night shift extrication waits on a second responder called from the village

The Bindarra night shift was rostered with one qualified mines rescue paramedic, based at the surface medical facility. The second responder, who must be present before any extrication is begun, was called in from the accommodation village 11 km from the gatehouse. That person travelled in the second response vehicle, which was garaged at the village and carried the hydraulic rescue tool. The post-incident review measured the extrication delay at approximately 9 minutes. The delay ran from 02:58, when the paramedic's primary survey was complete and the extrication plan made, to the second responder's arrival at 03:07.

The delay did not affect this occurrence or its outcome. The collision and the injuries were complete before the emergency response was paged. The driver was conscious, stable, restrained and protected by an intact roll-over structure throughout.

It is very likely that the arrangement increased risk all the same. On night shift, the site's extrication capability was a single person plus a drive from the village. No part of the arrangement changed with the severity of the casualty it was called to. Change one circumstance, such as a fuel fire, an unconscious casualty, or an airway that needed the vehicle opened in the first ten minutes, and the same wait would be a different outcome.

No safety action has been recorded against this issue.

Distribution

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